# Manasés Lovera

**AI Engineer · Machine Learning · Educator**

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## PROFESSIONAL SUMMARY

Versatile AI Engineer and Software Developer with hands-on expertise across the full AI product lifecycle — from fine-tuning deep learning models and building RAG pipelines to deploy production-grade agentic systems with MCP servers. Experienced in shipping AI-driven products at Tagshelf (computer vision, NLP, intelligent automation) and in educating the next generation of AI practitioners at Alterna Academy. Proficient in leveraging AI as a force-multiplier in software development workflows using LLM-assisted coding, prompt engineering, and AI pair-programming tools.

## EXPERIENCE

### AI / Software Engineer · Tagshelf

Nov 2024 – Present

*Santo Domingo, DO (Hybrid)*

- Architect and maintain Alfred, a production AI platform combining Computer Vision, NLP, and intelligent document processing, serving enterprise clients on a scale.
- Design and implement multi-step AI agent pipelines — orchestrating LLM reasoning, tool use, and retrieval-augmented generation (RAG) to automate complex business workflows.
- Build and expose MCP (Model Context Protocol) servers, enabling seamless integration of internal tools with LLM-based agents.
- Train, fine-tune, and evaluate deep learning models (CNNs, transformers) for document classification, OCR post-processing, and entity extraction.
- Lead adoption of AI-assisted development practices (GitHub Copilot, Claude, Cursor), reducing feature delivery time while maintaining high code quality.
- Implement scalable backend services with ASP.NET, FastAPI, and Express; deploy and monitor workloads on Azure and GCP.

AI Agents · RAG · MCP Servers · Fine-tuning · Computer Vision · NLP · Azure · GCP · ASP.NET · FastAPI

### AI Instructor · Alterna Academy

Sep 2025 – Present

*Santo Domingo, DO*

- Design and deliver a comprehensive AI curriculum: supervised/unsupervised learning, reinforcement learning, deep learning architectures, and LLM fundamentals.
- Teach hands-on modules on building AI agents, prompt engineering, agentic workflows, MCP integrations, and the Anthropic AI SDK.
- Guide students in building end-to-end ML projects: data preprocessing, model training with PyTorch & scikit-learn, evaluation, and deployment to cloud (Vertex AI, Azure ML).
- Mentor students on emerging topics: RAG pipelines, vector databases, agentic AI patterns, and modular AI system design.

Teaching · LLMs · AI Agents · PyTorch · RAG · Hugging Face · Azure ML · Vertex AI · MCP

### Software Developer · Oriontek

Apr 2024 – Jul 2024

*Santo Domingo, DO*

- Developed and deployed full-stack web applications using React and ASP.NET Core; managed cloud infrastructure on Azure.
- Collaborated on frontend integration with backend APIs, improving overall system performance and user experience.

React · ASP.NET Core · Azure

## TECHNICAL SKILLS

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|---|---|
| **AI / ML** | AI Agents · RAG Pipelines · MCP Servers · LLM Fine-tuning · Prompt Engineering · Computer Vision · NLP · Deep Learning · Reinforcement Learning |
| **Frameworks** | PyTorch · scikit-learn · Hugging Face · LangChain · Anthropic SDK · FastAPI · Flask · ASP.NET Core · NestJS · Express |
| **Cloud & MLOps** | Azure Machine Learning · GCP Vertex AI · Docker · GitHub Actions · Model deployment & monitoring |
| **Frontend** | React · Next.js · TypeScript · JavaScript · React Native · Flutter |
| **Databases** | PostgreSQL · SQL Server · Oracle · MongoDB |
| **Dev Tools** | GitHub Copilot · Claude · Cursor · Git · Linux · Colab · Codex · Antigravity |

## EDUCATION

**B.Sc. Software Engineering · Universidad Abierta Para Adultos (UAPA)** 2021 – Present

Full SDLC, software architecture, algorithms, and engineering best practices.

**Artificial Intelligence Specialization · Talento Digital** Nov 2025 – Jan 2026

ML model development from scratch, deep learning math, Azure ML, Vertex AI, PyTorch, scikit-learn, Colab.

**Full-Stack Development Bootcamp · Alterna Academy** Feb 2024 – Jun 2024

React, React Native, ASP.NET, cloud computing, Docker, git, Linux.

## LANGUAGES

**Spanish** Native · English Professional Working Proficiency

## REFERENCES

**Marlyn Pimentel** — Lead Software Engineer, Tagshelf · +1 809 780 9248
**María Rodríguez** — Operations Manager, Tagshelf · +1 829 380 7478